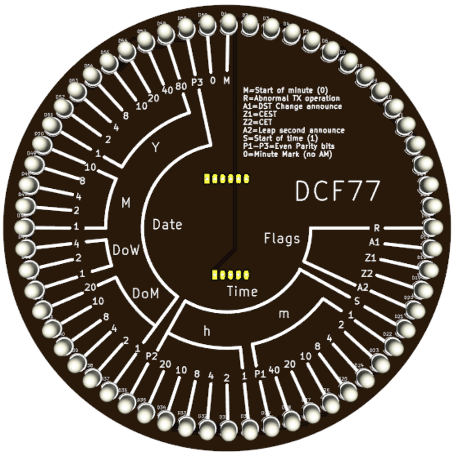
TD-FDCF01A



User Manual

REV 20260320

General

Features

The TD-FDCF01A is a clock showing time with 60 RGB LEDs that represent the bitstream of a DCF77 time signal. The clock can display the current time, in 24h format, the full date, and indicator flags showing summer/winter time being in use or being about to change. Thanks to clear markings on the silkscreen for each LED, it’s possible to easily read the time, at a glance, provided you know what you are looking at of course. This can be done without the need to memorize the bits position or refer to the manual.

The TD-M5037 movement gives this clock the ability to keep track of time even when the main power is removed. Additionally, the movement can automatically switch from/to DST (EU rules only), manual switchover is also possible, see TD-M5037 User Manual for all details.

NOTE: this is NOT a radio controlled clock. Time and date need to be set manually.

About the DCF77 Signal

The DCF77 is a time station, located near Frankfurt, that broadcasts a signal encoded with a bitstream representing the current time, date, and some additional information. The signal can be received all over Europe and is used by clocks to date and time sync automatically.
A DCF77 bit stream consists of 60 bits that are encoded on the transmitted carrier by reducing the power of the carrier for 100 or 200 mS at the start of each second. The duration of the power reduction signals a zero (100mS) or a one (200mS). Below is an explanation of the meaning of each bit.

Bit 0

Minute start. This bit is always zero.

Bit 1-14

German "Federal Office of Civil protection and Disaster Relief" Civil warnings. Additionally these same bits encode the current weather in several cities around Europe. The weather info is broadcast for a different city every minute and is encrypted. **NOTE:** because TD-FDCF01A does NOT receive a DCF signal these bits bear no real-world meaning and on the face they are generally off (save for the “seconds hand” going around or other effects the user might configure or program).

These bits CANNOT be used to determine the weather or civil alerts in this clock face as there is no DCF receiver.

Bit 15
“R” bit, if set, indicates an abnormal operation of the transmitter, always zero in this clock.

Bit 16
“A1” Summer time/Standard time change warning. Set one hour before a change in/out of DST occurs.

Bit 17
“Z1” Set when CEST is in use (Summer Time).

Bit 18
“Z2” Set when CET is in use (Standard Time).

Bit 19

"A2" Set during the hour before a leap second. Note: current firmware implementation doesn't calculate this bit.

Bit 20

Start of time, this bit is always set.

Bit 21-27

Current time minutes. Starting from Bit 21 each one has the following weight: 1,2,4,8,10,20,40 minutes. So, for instance, if Bit 21 and 27 are set the minutes are 1+40=41.

Bit 28

Minutes parity bit. This bit is an **even parity** bit over the bits 21-27. In other words, it's set or clear so that the amount of bits set in the whole 21-28 range is even. For instance consider this sequence:

Bit 21	Bit 22	Bit 23	Bit 24	Bit 25	Bit 26	Bit 27	Bit 28
1	0	0	0	0	1	1	1

Bits 21-27 have a total of 3 bits set, for this reason bit 28 is set so that the total amount of bits set is an even number (4). Conversely in this sequence:

Bit 21	Bit 22	Bit 23	Bit 24	Bit 25	Bit 26	Bit 27	Bit 28
1	0	0	0	0	1	0	0

Bits 21-27 have a total of 2 bits set, for this reason bit 28 is not set as the amount of set bits is already an even number.

Bit 29-34

Current time hours. Starting from Bit 29 each one has the following weight: 1,2,4,8,10,20 hours. So, for instance, if Bit 29 and 35 are set the hours are 1+20=21.

Bit 35

Hours parity bit. This bit is an **even parity** bit over the bits 29-34. In other words, it’s set or clear so that the amount of bits set in the whole 29-35 range is even. This works on the exact same principle as the minute bits.

Bit 36-41

Current day of the month. Starting from Bit 36 each one has the following weight: 1,2,4,8,10,20. So, for instance, if Bit 36 and 37 are set the day of the month is 1+2=3.

Bit 42-44

Current day of the week, with 1 being Monday and 7 Sunday. Starting from Bit 42 each one has the following weight: 1,2,4. So, for instance, if Bit 42 and 43 are set the day of the week is 1+2=3 which is Wednesday.

Bit 45-49

Current month with 1 being January and 12 December. Starting from Bit 45 each one has the following weight: 1,2,4,8,10. So, for instance, if Bit 45 and 46 are set the month is 1+2=3 which is March.

Bit 50-57

Current year in the century (0-99). Starting from Bit 50 each one has the following weight: 1,2,4,8,10,20,40,80. So, for instance, if Bit 50,52 and 55 are set the year is 1+4+20=25, so 2025.

Bit 58

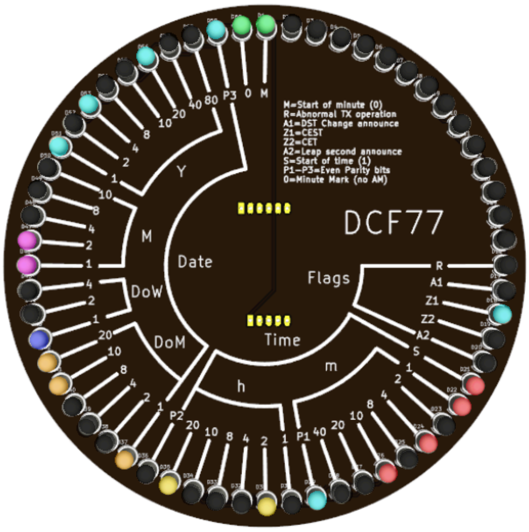
Date parity bit. This bit is an **even parity** bit over the bits 36-57. In other words, it's set or clear so that the amount of bits set in the whole 36-58 range is even. This works on the exact same principle as the minute and hours bits.

Bit 59

Minute Marker. This is the only bit in the whole stream that is never modulated, which means the amplitude is constantly 100% (neither a zero nor a one). This is used by the receivers to detect the beginning of the stream as the next bit.

Reading Time

With the information above you should be able to read the date and the time from the clock face. Below is an example that we will use to clarify how the clock is read.



If you like the challenge, write down the date and time you can read before moving on to the next paragraph.

Starting from the zero position the first LED we find lit is the Z1 indicator. As seen also on the PCB silkscreen, Z1 indicates CEST is in use (Central European Summer time). Next is the S marker, this is the start of time and is always on, so we can ignore it.

Under the “**m**” (minute) bracket we have 3 LEDs lit, at positions 1,4 and 10. So, the minutes are 10+4+1=**15**. Note that P1 is also lit. This is because the “m” section has 3 LEDs on, so the parity P1 brings these to an even number (4).

Next are the hours, under the “**h**” bracket. Here we have 2 LEDs lit: 2 and 20, so the hours are 20+2=**22**. Note that P2 is off because the “h” section has already an even number of LEDs lit (2).

The following section, “**DoM**” is the day of the month. As above here 1,10,20 are lit so the day is 1+10+20=**31**.

Next is “**DoW**”, day of the week. With only 1 lit this is a **Monday** (Sunday is 7).

“**M**”, month is 2+1=3 so **March**

The year, “**Y**”, is 1+4+20=25 which, in the current century, is interpreted as **2025**.

Finally, if you count all the LEDs set under the bracket “Date” they are 9. This is an odd number so P3 (which refers to the all set of bits in the date) is lit to make the total 10, an even number.

So, in summary, the face is showing: **Monday, 31 March 2025** and time is **22:15**. We also learn that we are on summer time.

Setting the Time

Time set mode is entered with a long press on the A button. The first settable value (the hours) will start to flash on the clock face. Additionally the LED at zero position will glow steadily instead of flashing seconds.

NOTE: Upon entering time set mode you will not see any lights flashing if the current hours are "0". Just press B or C to increase/decrease and you will see some values flashing.

Once time set mode is entered use B and C to increase/decrease the value being set respectively. To move to the next value (minutes) just press once A. Pressing again the A button will move you to the next settable value. Continue like this for the day, day of week (remember 1 is Monday and 7 is Sunday), month and year. Once you are done setting the time, long press again A to exit time set mode.

Hint: to achieve a second-accurate setting, set the clock to the next minute. Then wait in time set mode until your reference clock shows :59 seconds, only at this point long press on A to exit time set mode. This works because, upon exiting time set mode, the clock always starts from second :00.

Display Test Mode

It’s possible to enter test mode by long pressing on the C button. In this mode your clock will light all LEDs regardless of current time. This can be useful to ensure all LEDs are functioning and to test your power supply and ensure it can provide enough current. Long press again C to exit test mode.

Display Modes

The face supports 4 different colour palettes. Out of the box these are 3 different luminosity levels and, the 4th, is all LEDs red. The colour scheme can be changed by using the serial interface (see movement “Advanced” section for commands to set registers). A detailed memory map of the colour registers for this face can be found in the “Advanced” section of this manual.

During normal operation you can cycle through the modes by clicking the **A** button.

Assembling the Board

Packing List

Before proceeding with the assembly of the components on the board, double check the packing list to make sure you have received all components and that everything is in good order.

60 x 5mm WS2812D LEDs
1 x Pin 1x5 Vertical Male 1 x Pin 1x6 Vertical Male 3 x Tactile Push Buttons
1 x PCB TD-FDCF01A REV.A

Top Assembly

NOTE: If you decided to use the printed plastic with the slits to shine light on the PCB white silkscreen you will need to bend the LED pins 90 degrees very close to the body so that the LED points towards the center of the clock and still has the flat surface aligned with the silk screen. I strongly recommend bending a few LEDs, pushing them in without soldering on different sides of the ring and making sure the plastic top fits. If you chose to not use the cover plastic ring, then the LEDs should be mounted facing upwards.

Insert all LEDs paying attention to orient the LED flat side as indicated by the silk screen. Ensure LEDs are all sitting at the same height. Notice how the LED pads are staggered (see picture below).

This facilitates manual soldering, given the relatively small clearance between the pads, it would otherwise be quite easy to create solder bridges. I found that the most convenient way to solder the LEDs is to put in a few and then solder first the square pins using a wider gauge tin (0.8mm worked very well for me). Next flip the board and make sure all LEDs are vertical, at the same height and generally well aligned. If something is off it's very easy at this stage to rework it by just heating the single soldered pad holding them in place.

Once happy with the alignment, solder the pins opposite the square ones using the same 0.8mm tin. When this is done, before doing the two middle pins, it helps to clip all the terminals close to the board. In this way it's possible to solder the middle pins working from top, and not the side of the pin, minimizing the chances of a bridge forming. For this step it’s best to use a smaller gauge tin (for instance 0.2mm, but you might find other sizes working better depending on your solder tip).

Bottom Assembly

Once all the LEDs are in place, you can add the pin strips that allow the movement to be plugged in. Make sure they are installed on the opposite side of the LEDs, as per silk screen.

You can finally then solder the 3 push buttons.

There are no other components on this PCB.

Testing

Power up the clock and enter test mode by long pressing the C button. This will cause all LEDs to light up ensuring everything is soldered correctly. You can exit test mode by long pressing on C again. You can now set the time so you ensure all other buttons are working and the movement is operating as expected. Both in Test Mode and in regular mode you can change the intensity of the LEDs by pressing on A, this will cycle through 3 levels and OFF screen mode.

If you have decided to use the printed plastic with the slits to shine light on the PCB, you should place it on top of the PCB in test mode with all lights on. Then slightly rotate the plastic to align the slits with LED/markings on the PCB, till you find a position with maximum light. At this point you should fix the plastic in place with a few small drops of hot glue at the back to keep it in place.

Advanced

Your clock comes with a UART that you can use to send commands to it via a terminal program. This is not only an alternative way to set time, it's also a way to change the clock colours. See the TD-M5037 User Manual for details on how to connect, send commands and alter register values.

See next page for a detailed registers' map of this clock face.

0x00-0x3F Movement Registers	See TD-M5037 User Manual for usage.	
0x40-0x7F Face Registers	0x40-0x42	Colour of OFF LEDs in R,G,B
	0x43-0x45	Colour of Minutes LEDs in R,G,B
	0x46-0x48	Colour of Hours LEDs in R,G,B
	0x49-0x4B	Colour of DoM LEDs in R,G,B
	0x4C-0x4E	Colour of DoW LEDs in R,G,B
	0x4F - 0x51	Colour of Month LEDs in R,G,B
	0x52-0x54	Colour of Year LEDs in R,G,B
	0x55-0x57	Colour of Flags LEDs in R,G,B
	0x58-0x5A	Colour of Markers LEDs (start of time, start of minute, minute marker) in R,G,B
	0x5B-0x5D	Colour of Civil Warning Bits LEDs in R,G,B (currently unused, used only in

		test mode to show LEDs are working)
	0x5E-0x60	Colour of Seconds Hand LEDs in R,G,B Note: as the seconds' hand moves around the clock it temporarily shadows the bit of the current second. You can disable the seconds' hand by setting these registers to 0x000000.
	0x61-0x6F	Unused, colour table padding.
	0x70-0x9F	Second color table, bytes meaning same as the first section above. This table can be activated by pressing A once.
	0xA0-0xCF	Third color table, bytes meaning same as the first section above. This table can be activated by pressing A once more.
	0xD0-0xFF	Fourth color table, bytes meaning same as the first section

		above. This table can be activated by pressing A once more.

Appendix A - HW Revisions

REV.A	First version.